Posted on 29 Mar 2025 — The copyright holder is the author/funder. All rights reserved. No reuse without permission. — https://doi.org/10.22541/essoar.174326513.30629624/v1 — This is a preprint and has not been peer-reviewed. Data may be preliminary.

Use of Double Cole Cole Model for The Estimation of Pore Size Distribution of Carbonates

Neha Panwar[1]

[1]Affiliation not available

March 29, 2025

(1) Indian Institute of Technology, Roorkee, Uttarakhand, India

Corresponding Author: Ravi Sharma (ravi.sharma@es.iitr.ac.in)

Department of Earth Sciences, IIT Roorkee, Roorkee, Uttarakhand,247667

Emails: npanwar@es.iitr.ac.in; hjyotikalita@es.iitr.ac.in; ashish_d@es.iitr.ac.in; priteshi-itr@gmail.com;smalik@es.iitr.ac.in; ppandey2@es.iitr.ac.in

Running title: Characterizing Pore Size Distribution in Carbonates

Summary.

Estimating storage and flow properties in carbonate reservoirs presents challenges due to their complex porosity and permeability distribution. This study investigates the potential of spectral induced polarization (SIP) as a non-destructive method for analyzing pore size distribution (PSD) in carbonate formations. To validate the SIP results, traditional techniques such as mercury intrusion porosimetry (MIP) and scanning electron microscopy (SEM) image processing were utilized. Additionally, a modified open-source MATLAB code for relaxation time distribution (RTD) was employed to invert SIP spectra, which were then fitted with a Double Cole-Cole model. The results obtained from SIP closely aligned with those from MIP, affirming the reliability of SIP for PSD analysis. Furthermore, advanced image processing techniques, including the watershed algorithm, were applied to SEM images to further analyze PSD. The integration of SIP, MIP, and SEM techniques provided valuable insights into microporosity in carbonates, revealing a broad grain size distribution encompassing fine-grained carbonate mud and larger biogenic fragments. This study establishes a clear correlation between electric behavior (quadrature conductivity) and petrophysical properties (grain and pore size distribution, storage, and permeability), thereby enhancing the predictive modeling of reservoir properties in complex carbonate rock systems.

Keywords: Complex conductivity , Grain size distribution (GSD), Relaxation time distribution (RTD), Image processing, Rock physics, Double Cole-Cole model

1. Introduction

Carbonate rocks exhibit heterogeneity in their textures and properties, resulting in a varied distribution of storage and flow properties. These rocks are significant for the storage of oil, gas, and water and are responsible for more than 60% of oil production and 40% of natural gas production globally (Kargarpour, 2020; Weiqiang et al., 2020). Carbonate reservoirs are the host for some of the largest fresh groundwater storage (Cunningham et al., 2012; Kovács, 2021). Due to the diagenetic process, referring to the physical and chemical changes after deposition, these rocks exhibit mineralogical, structural, and textural heterogeneity (Lu et al., 2021) at all scales i.e., 10μm – 10 mm, 1cm-10cm, 10cm-10m, 10m-1000m (Brown, 2011). Despite the importance of carbonate rocks, there remains a significant gap in the literature regarding the accurate

Posted on 29 Mar 2025 — The copyright holder is the author/funder. All rights reserved. No reuse without permission. — https://doi.org/10.22541/essoar.174326513.30629624/v1 — This is a preprint and has not been peer-reviewed. Data may be preliminary.

characterization of pore size distribution (PSD) and its implications for fluid flow and storage. The porosity in carbonates has been classified into different types based on deposition (Choquette and Pray, 1970), relation of petrophysical properties with rock fabric (Lucia, 1983, 1995) and a combination of deposition and rock fabric (Lonoy, 2006). Under these dynamic scenarios, a realistic estimate of the petrophysical properties and other related behaviors in the carbonate rocks assumes greater significance (Bust et al.,2011). Many researchers (Revil et al., 2012; Revil et al., 2014; Cao et al., 2016; Zhang et al., 2022) have attempted to tackle the challenge of determining pore size distribution (PSD) in siliciclastic rocks by employing numerous laboratory methods and sometimes even performed an integrated analysis of two or more techniques for convergence and reliability. Among the many available laboratory methods to determine the PSD, mercury intrusion porosimetry (MIP) (Cnudde et al., 2009; Sun et al., 2020, Bohnsack et al., 2021), and nuclear magnetic resonance (NMR) (Zheng et al., 2020; Hussein et al., 2018; Niu & Zhang, 2018) are the two most established technique and have gained wide acceptability as the industry standard methods to determine PSD. However, both methods have associated limitations that can affect their applicability. MIP is a destructive technique that results in the loss of samples, while NMR requires extensive sample preparation, making it costly and time-consuming to use. For instance, when estimating PSD using NMR-derived relaxation time distribution (RTD), the assumption of a homogeneous distribution of pores can lead to an overestimation of large pore sizes (Stingaciu et al., 2010). In addition to MIP and NMR, scanning electron microscopy (SEM) image processing is commonly employed to estimate pore size distribution (PSD), with a particular focus on pore body size, across various formations of interest, including sandstones, shales, and carbonate (Diamond & Leeman, 1994; Beiranvand, 2003; Widiatmoko et al., 2010; Yao et al., 2014; Zhang et al., 2018; Liu et al., 2020; Lu et al., 2021; Safari et al., 2021). SEM provides high-resolution images that allow for detailed analysis of pore structures, complementing the data obtained from MIP and NMR. Various methods, including μ-CT, MIP, NMR, and SIP, have been employed to analyze pore size distributions in sandstone samples, each utilizing distinct methodologies. This diversity in approaches underscores the importance of integrating multiple techniques to achieve a comprehensive understanding of the pore structures present in these samples. Muller-Huber et al., (2018) used the Cole-Cole model in grain-dominated and mud-dominated carbonate rocks to derive the Cole-Cole parameters (c, τ, m) and compared the outcomes with NMR and MIP data. They found that the grainstones have high value of Cole-Cole exponent (c) whereas mudstones have a low value of c. Zhang et al., (2018) estimated PSD using joint inversion of SIP and NMR data of one oomoldic dolostone sample and a mismatch between the joint inversion and NMR data was found.

The SIP method measures the bulk polarization effects during the relaxation of ions over a range of frequencies (typically ranging from 0.1-1000 Hz), once the static current is taken off. Spectral response is generated by electrical double-layer polarization formed at the mineral fluid interface. At lower frequencies (<100 Hz), both electric double-layer polarization and membrane polarization can contribute significantly to the total polarization response. The electric double-layer polarization is typically the dominant mechanism, but the contribution from membrane polarization can also be important, depending on the specific mineral-fluid system being studied and the characteristics of the pore space. At higher frequencies (>1 kHz), other effects such as Maxwell-Wagner polarization and dielectric polarization may start to mask the measured conductivity (Revil et al., 2017). The Maxwell-Wagner polarization occurs due to the interfacial charge accumulation when two different dielectric materials incorporate a complex membrane at the heterogeneous interface (Li et al., 2021). If the stern layer polarization (EDL) is the primary polarization mechanism at lower frequencies, it is believed that the relaxation time distribution is associated with the PSD obtained from the inverted polarization measurements (Tong et al., 2006; Revil & Florsch, 2010; Revil et al., 2014, Jougnot et al., 2010). The Cole-Cole and Double Cole-Cole model are used to analyze complex conductivity data based on the relaxation time distribution (Pelton et al.,1978).

In this paper, we aim to investigate the use of the SIP method for characterizing the PSD of carbonate rocks using the Double Cole-Cole model. This study addresses the existing gap in the literature regarding the integration of SIP with established methods such as MIP and SEM image processing, which remains largely unexplored. While several studies have explored PSD characterization of carbonate rocks through various laboratory techniques, the integration of SIP with established methods such as MIP and SEM image process-

Posted on 29 Mar 2025 — The copyright holder is the author/funder. All rights reserved. No reuse without permission. — https://doi.org/10.22541/essoar.174326513.30629624/v1 — This is a preprint and has not been peer-reviewed. Data may be preliminary.

ing remains largely unexplored. Previous studies have typically focused on one or two carbonate samples, limiting the scope of their findings. By employing a larger dataset and combining SIP with complementary methods, we seek to achieve a more accurate and comprehensive estimation of the petrophysical properties of carbonate rocks, thereby advancing the understanding of their behavior in geological and reservoir contexts.

2. Theory:

The determination of PSD in rocks is a crucial aspect in comprehending their physical and chemical characteristics. Various techniques, such as SIP, MIP, and SEM, can be employed to estimate the pore size distribution.

2.1 SIP and its inversion:

SIP/Complex conductivity is a non-invasive geophysical technique that is sensitive to the size, shape, and distribution of the pores in the rock matrix. It provides valuable information about the electrical properties of the rock, particularly related to the pore characteristics, such as the bulk conductivity, surface conductivity, and the electrical double layer (EDL) of the mineral surfaces within the rock matrix. It encompasses both the conduction and polarization mechanisms and is expressed in terms of in-phase conductivity and quadrature conductivity. Equation 1&2 describes the in-phase and quadrature conductivity, respectively (Halisch et al., 2014; Osterman et al., 2016).

$\sigma^{'} = \frac{1}{F}(\sigma_w + \sigma_s^{'})$ (1)

$\sigma^{"} = \sigma_s^{"}$ (2)

Where, σ_w is the pore water conductivity, F is formation factor, $\sigma_s^{'}$ and $\sigma_s^{"}$ are the real and imaginary part of the surface conductivity, respectively.

The Cole-Cole model is a complex mathematical model that describes the complex conductivity of a material as a function of frequency. It assumes that the electrical properties of the material can be represented by a distribution of relaxation times, which correspond to the time scales over which charges move in response to an applied electric field. The model comprises three parameters such as frequency-dependent electrical conductivity, a relaxation time, and an exponent that describes the broadness of the relaxation spectrum. The relaxation time arises when the ions in EDL move back to their original position after the current is switched off and is defined by equation 3 (Vinegar,1984; Revil et al., 2012; Liu et al., 2014; Coperey et al., 2019).

$\tau_0 = \frac{r^2}{aD_i}$ (3)

Where, τ_0 is the relaxation time constant (seconds), r is pore radius (μm), a is constant and D_i is the diffusion coefficient. Determining the relaxation time from complex conductivity data is commonly achieved by fitting experimental data with empirical relations such as the Cole-Cole or Double Cole-Cole models etc.

In this section, we describe the theoretical basis and model employed in the code adapted from Florsch et al., (2014). This adaptation is crucial for our analysis of spectral induced polarization (SIP) data, as it allows us to effectively estimate pore size distributions (PSD) in carbonate rocks. We detail the underlying principles and equations utilized in the original code, considering any assumptions or simplifications made by the authors. Our approach to SIP data inversion involves the integration of two distinct disciplines. Initially, a prior guess is essential to initiate the inversion process. This prior guess cannot be arbitrary; rather, it must be grounded in a fundamental or elementary model. Subsequently, this prior guess or model serves as the starting value (initial guess) within the inversion framework.

To derive the equation which will be used for getting initial guess or prior guess for time-varying generalized least squares (TVGLS) inversion (Tarantola & Valette, 1982), we consider the following elementary model which is a special case of generalized linear admittance or impedance model, when only one type of pore size is present having one relaxation time τ_0 is given by equation 4.

$\sigma^* = \sigma_\infty + \frac{\sigma_0 - \sigma_\infty}{1 + i\omega\tau_0} + i\omega C$ (4)

Posted on 29 Mar 2025 — The copyright holder is the author/funder. All rights reserved. No reuse without permission. — https://doi.org/10.22541/essoar.174326513.30629624/v1 — This is a preprint and has not been peer-reviewed. Data may be preliminary.

Generally, a range of pore sizes corresponds to a distribution of relaxation times (?), described by the relaxation time function ?(?). If we consider this distribution to be normally distributed, then

$\int_0^\infty g_\tau(\tau)d\tau = 1$

where RTD, g is normalized.

σ_0 and $\sigma_{[?]}$ are DC and instantaneous conductivity, τ is relaxation time, C is a constant having the dimension of a capacitance and used to model the high frequency trend of the spectrum. To get a generalized linear admittance or impedance model (equation 6), equation 4 is changed by the following steps (Florsch et al., 2014):

$$a_R = \sigma_\infty$$

$M = -(\sigma_0 - \sigma_\infty) \geq 0$ (5)

$$g_\tau(\tau) = \delta(\tau_0 - \tau)$$

$$\varphi(\omega, \tau) = 1/(1 + i\omega\tau)$$

$$\sigma^* = \sigma_\infty + (\sigma_0 - \sigma_\infty).\frac{1}{1 + i\omega\tau_0} + i\omega C$$

$\int_0^\infty g_\tau(\tau)d\tau = 1 \& g_\tau(\tau) = \delta(\tau_0 - \tau)$

So, $\int_0^\infty g_\tau(\tau)\delta\tau$ will be 1 when $\tau = \tau_0$

$$\sigma^* = \sigma_\infty + (\sigma_0 - \sigma_\infty).\frac{\int_0^\infty g_\tau(\tau)\mathrm{d}}{1 + i\omega\tau_0} + i\omega C$$

$$\sigma^* = \sigma_\infty + (\sigma_0 - \sigma_\infty).\int_0^\infty \frac{g_\tau(\tau)\,\mathrm{d}}{1 + i\omega\tau} + i\omega C$$

$\sigma^* = a_R - M.\int_0^\infty g_\tau(\tau)\,\varphi(\omega, \tau)d\tau + i\omega C$(6)

Where, a_R is real constant, M is a constant indicating the amplitude of induced polarization (IP), $g_\tau(\tau)$ is relaxation time distribution (RTD), $\varphi(\omega, \tau)$ is the kernel.

Now in managing equation 6 we have to determine both the constant M and RTD with additional condition that it must be normalized, it is however simpler to encapsulate M into $g_\tau(\tau)$ and write the model in the form of equation 7.

$\sigma^* = a_R - \int_0^\infty g_\tau(\tau)\,\varphi(\omega, \tau)d\tau + i\omega C$(7)

In the following we consider that M has been encapsulated in RTD and recovery of the true distribution requires normalization. Now usual change of variable is performed which better accommodates frequency scale invariance:

$$z = -\log(\omega) \leftrightarrow \omega = e^{-z}$$

$$s = \log(\tau) \leftrightarrow \tau = e^{s}$$

Posted on 29 Mar 2025 — The copyright holder is the author/funder. All rights reserved. No reuse without permission. — https://doi.org/10.22541/essoar.174326513.30629624/v1 — This is a preprint and has not been peer-reviewed. Data may be preliminary.

The above change of variables yields equation 8:

$A_z(z) = a_R - \int_{-\infty}^{\infty} G_s(s) \varnothing(z,s)\, ds + ie^{-z}C(8)$

Where

$$G_s(s) = \tau g(\tau)$$

$$\varnothing(z,s) = \varphi(e^{-z}, e^{s})$$

Now considering the homogenous case in which distribution is assumed to be constant on given interval that is:

$$G_s(s) = \tau g(\tau) = constant = G_0$$

Which gives us the equation 9 to be used for deriving initial or prior guess by putting above relation in equation 7:

$\sigma^* = a_R - \int_{T_{\min}}^{T_{\max}} \frac{1}{\tau} \frac{G_0}{1+i\omega\tau} d\tau + i\omega C(9)$

Integrating the equation 9 and separating the real and imaginary parts gives us the d = Gm format equation which is further used for inversion to get the prior or initial guess model parameters.

TVGLS is followed for inversion in which the equation 7 is first discretized using the Riemann sum method. In order to get positive relaxation time logarithmic change of variable is done, and real and imaginary part of equation are separated to get the d = Gm form.

In light of the intricate dynamics characterizing the relaxation process of carbonates, a pivotal adjustment was made to the code. Specifically, the introduction of the Double Cole Cole model (as delineated in Equation 10) was implemented to effectively accommodate the spectral induced polarization (SIP) data and facilitate the inversion of relaxation time distributions relevant to carbonates. The Double Cole Cole model is an extension of Cole-Cole model in terms of relaxation times given as:

$\sigma_\omega^* = \sigma_\infty \left[1 - \frac{M_1}{1+\iota\omega{\tau_1}^{c1}} - \frac{M_2}{1+\iota\omega{\tau_2}^{c2}}\right](10)$

Where M_1 and M_2 are the chargeability,τ_1 and τ_2 are the relaxation time constants, c_1 and c_2 are Cole-Cole time constants and $\omega = 2\pi f$, is the angular frequency.

The real and imaginary part of the Double Cole-Cole model is given in equation 11 and 12, respectively:

$\sigma' = \sigma_\infty[1 - \frac{M_1\omega{\tau_1}^{c1}(c_1\cos\frac{\pi}{2})}{(1+\omega{\tau_1}^{c1}(c_1\cos\frac{\pi}{2})^2+(\omega{\tau_1}^{c1}\sin\frac{\pi}{2}c_1)^2} - \frac{M_2\omega{\tau_2}^{c2}(c_2\cos\frac{\pi}{2})}{(1+\omega{\tau_2}^{c2}(c_2\cos\frac{\pi}{2})^2+(\omega{\tau_2}^{c2}\sin\frac{\pi}{2}c_2)^2}]$ (11)

$\sigma'' = \sigma_\infty[\frac{M_1\omega{\tau_1}^{c1}(c_1\sin\frac{\pi}{2})}{(1+\omega{\tau_1}^{c1}(c_1\cos\frac{\pi}{2})^2+(\omega{\tau_1}^{c1}\sin\frac{\pi}{2}c_1)^2} - \frac{M_2\omega{\tau_2}^{c2}(c_2\sin\frac{\pi}{2})}{(1+\omega{\tau_2}^{c2}(c_2\cos\frac{\pi}{2})^2+(\omega{\tau_2}^{c2}\sin\frac{\pi}{2}c_2)^2}]$ (12)

The process to invert relaxation time distribution using Double Cole-Cole is explained in Appendix A.

By integrating this model into our code, we're positioned to gain detailed insights into how pore structure and electrical properties interact in carbonate reservoirs.

2.2 Mercury Intrusion Porosimetry (MIP):

For mercury intrusion porosimetry (MIP) measurements, the sample was analyzed using the PoreMaster 60 (Quanta Chrome Instruments). MIP was conducted to estimate the pore size distribution of the sample by determining the volume of mercury intruded at different pressure levels. The relationship between pressure and capillary diameter is governed by the Washburn equation 1 (Washburn, 1921; Abell et al., 1999):

$P = \frac{-4\gamma cos\theta}{d}$ (13)

Posted on 29 Mar 2025 — The copyright holder is the author/funder. All rights reserved. No reuse without permission. — https://doi.org/10.22541/essoar.174326513.30629624/v1 — This is a preprint and has not been peer-reviewed. Data may be preliminary.

Where, P = Pressure, = Surface tension of the liquid, ϑ = Contact angle of the liquid and d = diameter of the capillary.

2.3 Scanning Electron Microscopy (SEM):

SEM is another technique used in conjunction with other methods to estimate pore size distribution. SEM involves using an electron microscope to produce high-resolution images of the rock matrix, allowing for direct observation and measurement of pore size and shape. This method provides detailed information about the microstructure of the rock, complementing the data obtained from techniques like SIP and MIP.

By combining the results from SIP, MIP, and SEM, researchers can gain a comprehensive understanding of the pore size distribution in rocks. SIP provides insights into the complex conductivity influenced by pore characteristics, while MIP offers direct measurements of pore size based on mercury intrusion. SEM enhances the analysis by providing visual confirmation and detailed information on pore morphology. Together, these methods contribute to a more comprehensive assessment of the physical and chemical properties of rocks through the estimation of pore size distribution.

3. Material and Methodology:

This research involves 11 carbonate rock samples collected from the carbonate platforms of the Tertiary sequence, specifically from the middle Eocene to early Oligocene periods, offshore of Western India. The samples exhibit various features, including fractures, vugs, and channels. Notably, they possess intercrystalline porosity, which is formed in the spaces between dolomite crystals, as well as vuggy porosity, a type of secondary porosity characterized by pore spaces created by solution vugs (Irajian et al., 2016; Sharma et al., 2013). Additionally, some samples exhibit channel porosity, which consists of irregular, elongated pores that cut across the rock fabric and are typically associated with the solution enlargement of fractures. The porosity of the samples ranges from 0.07 to 0.22 (fraction), while the permeability varies between 0.00496 mD and 10.94 mD. These characteristics highlight the diverse porosity types present in the carbonate rock samples and their potential implications for fluid flow and storage.

3.1 SIP data acquisition:

Hosted file

`image2.emf` available at https://authorea.com/users/907675/articles/1281815-use-of-double-cole-cole-model-for-the-estimation-of-pore-size-distribution-of-carbonates

The SIP or complex conductivity measurements were performed on 11 carbonate core samples using the ZELSIP04-V02 impedance meter (Zimmermann et al., 2008) within a frequency range spanning from 100 mHz to 45 kHz, as illustrated in Figure 1. Prior to the measurements, the samples underwent drying in an oven at 60°C and subsequent saturation with tap water under vacuum conditions. This meticulous preparation ensured uniformity and minimized variability in moisture content across the samples, thereby enhancing the reliability and consistency of the SIP measurements. Non-polarizable current and voltage electrodes were utilized during the SIP measurements to mitigate polarization effects and facilitate accurate impedance measurements.

Figure 1: Experimental setup to conduct SIP measurements using ZEL impedance meter. The sample is connected to the ZEL impedance meter with current electrodes (A & B) and potential electrodes (M & N).

3.2 MIP data acquisition:

For MIP measurements, a quantity of 2g of sample was used in the Pore Master 60 ©Quanta chrome instrument. The instrument consists of two sections. The first section includes a low-pressure section used for evacuating the sample cells, filling them with mercury, and measuring the largest pore in the sample using pneumatic pressure below 50 psia. The second section includes hydraulic high pressure that performs intrusion measurements at pressure up to 60,000 psia.

3.3 SEM data acquisition:

Posted on 29 Mar 2025 — The copyright holder is the author/funder. All rights reserved. No reuse without permission. — https://doi.org/10.22541/essoar.174326513.30629624/v1 — This is a preprint and has not been peer-reviewed. Data may be preliminary.

We utilized a Carl Zeiss EVO 18 scanning electron microscope (SEM) for data acquisition, offering a magnification range from <5X to 1,000,000X and a resolution of up to 3072 × 2304 pixels. The sample preparation involved cleaning the stub with ethylene glycol and affixing a carbon tape onto it. The coated sample was then inserted into the SEM. Image acquisition was performed at resolutions of 1 μm, 2 μm, 10 μm, 20 μm, 100 μm, and 200 μm.

The acquired SEM images were processed to determine the pore throat size distribution using the following methodology: the image quality was first improved using a median filter, followed by segmentation using the Otsu threshold method. The watershed algorithm was then applied to the segmented images to extract the pore throats. The watershed algorithm constructs boundaries around the pores, known as watershed lines. This process can be visualized by assuming that water is injected from the bottom in a basin, and the moment water starts to spill into adjacent basins, a dam is constructed at the point of overflow. These dams define the pore throats connecting one pore to another pore (Gonzalez and Woods, 2018).

3.3.1 Grain size distribution estimation:

The grain size distribution was estimated using ImageJ software as explained below:

1. Firstly, the SEM data was translated into pixel dimensions to establish a scale for measurement, allowing the grains to be measured from end to end. This step is crucial for ensuring accurate size estimation relative to the actual dimensions of the grains.

2. Next, multiple measurements were recorded for each individual grain to account for variability in size. These measurements were then utilized to construct a histogram representation of the grain size distribution, providing a visual summary of the size range and frequency of grains within the sample.

4.Results & Discussion:

To analyze the relaxation behavior of carbonates, we employed an open-source MATLAB code to fit the experimental SIP data with empirical models such as Cole-Cole or Double Cole-Cole. Initially, our aim was to determine the most suitable model for explaining the relaxation behavior of carbonates. The Cole-Cole model was used to fit the experimental data, as shown in Figure 2(a) & (b). While the Cole-Cole model effectively matched the experimental data of the homogeneous texture samples, namely I14 and I15, it inaccurately estimated the relaxation time distribution (Figure 2c).

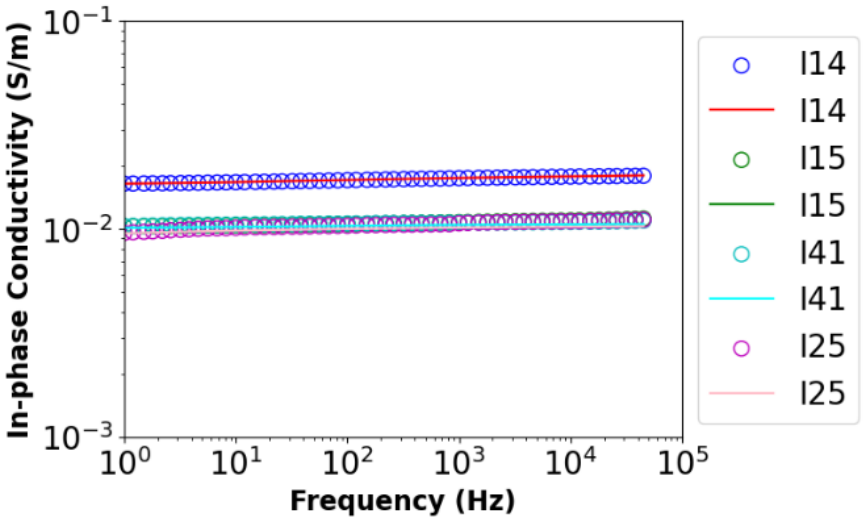


Figure 2(a &b): SIP spectra (a) In-phase and (b) Quadrature conductivity, of carbonate samples fitted with Cole-

Cole model.

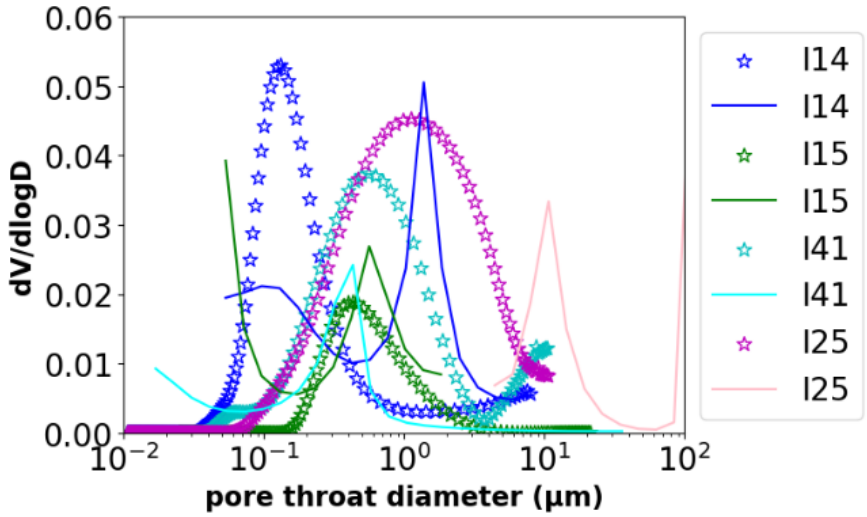


Figure 2(c): Pore size distribution of carbonate samples fitted with inverted SIP data. (*) represents the experimentally measured pore size distribution using MIP, while the line illustrates the estimated dominant pore size distribution obtained through analysis with the RTD code.

For the heterogeneous texture samples (I25 and I41), the Cole-Cole model failed to adequately fit the measured SIP data. This discrepancy highlights the limitations of the Cole-Cole model when applied to more complex pore structures. Furthermore, we observed that when the Cole-Cole exponent (c) had non-unique values, the results obtained from the Cole-Cole model were still unique. This suggests that the relaxation behavior of carbonates is inherently complex and cannot be fully explained by the Cole-Cole model alone. Therefore, more sophisticated models may be necessary to accurately represent the relaxation behavior of the material. The presence of dual porosity i.e., coexistence of two distinct types of pore systems within the rock, in our carbonate samples and hence complex relaxation behavior of the heterogeneous samples were confirmed through SEM images, as shown in Figure 3. The heterogeneous sample exhibits fracture porosity with dominant intracrystalline porosity, while homogeneous samples show uniform distribution of pores and grains. This observation reinforces the notion that the structural complexity of carbonate rocks significantly influences their relaxation behavior, necessitating the use of advanced modeling techniques for accurate characterization.

Posted on 29 Mar 2025 — The copyright holder is the author/funder. All rights reserved. No reuse without permission. — https://doi.org/10.22541/essoar.174326513.30629624/v1 — This is a preprint and has not been peer-reviewed. Data may be preliminary.

Posted on 29 Mar 2025 — The copyright holder is the author/funder. All rights reserved. No reuse without permission. — https://doi.org/10.22541/essoar.174326513.30629624/v1 — This is a preprint and has not been peer-reviewed. Data may be preliminary.

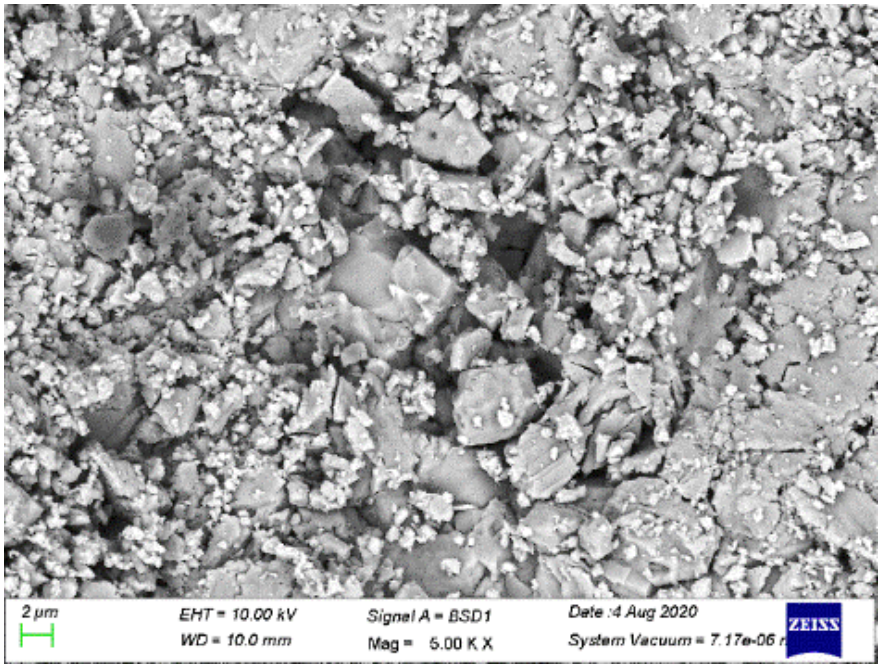


Figure 3: Back Scattered (BSE) SEM backscattered images of carbonate sample. I14 & I15 shows the uniform surface morphology, consistent grain sizes, and a lack of distinct variations emphasize the overall homogenous nature at same scale. I25 & I41 shows the Varied textures, grain sizes, and surface features highlight the diverse composition within the sample at same scale.

Following the failure of the Cole-Cole model in explaining the relaxation behavior of heterogeneous samples, we switched to the Double Cole-Cole model. The experimental SIP data matched excellently with the parameters of the Double Cole-Cole model. Figure 4 illustrates the fitting of the experimental SIP data using the Double Cole-Cole model. Table 1 presents the estimated values of the Double Cole-Cole parameter (M_1, M_2, c_1, c_2, τ_1, τ_2, $\sigma_{[?]}$), obtained by inverting the real and quadrature parts of the SIP spectra.

Posted on 29 Mar 2025 — The copyright holder is the author/funder. All rights reserved. No reuse without permission. — https://doi.org/10.22541/essoar.174326513.30629624/v1 — This is a preprint and has not been peer-reviewed. Data may be preliminary.

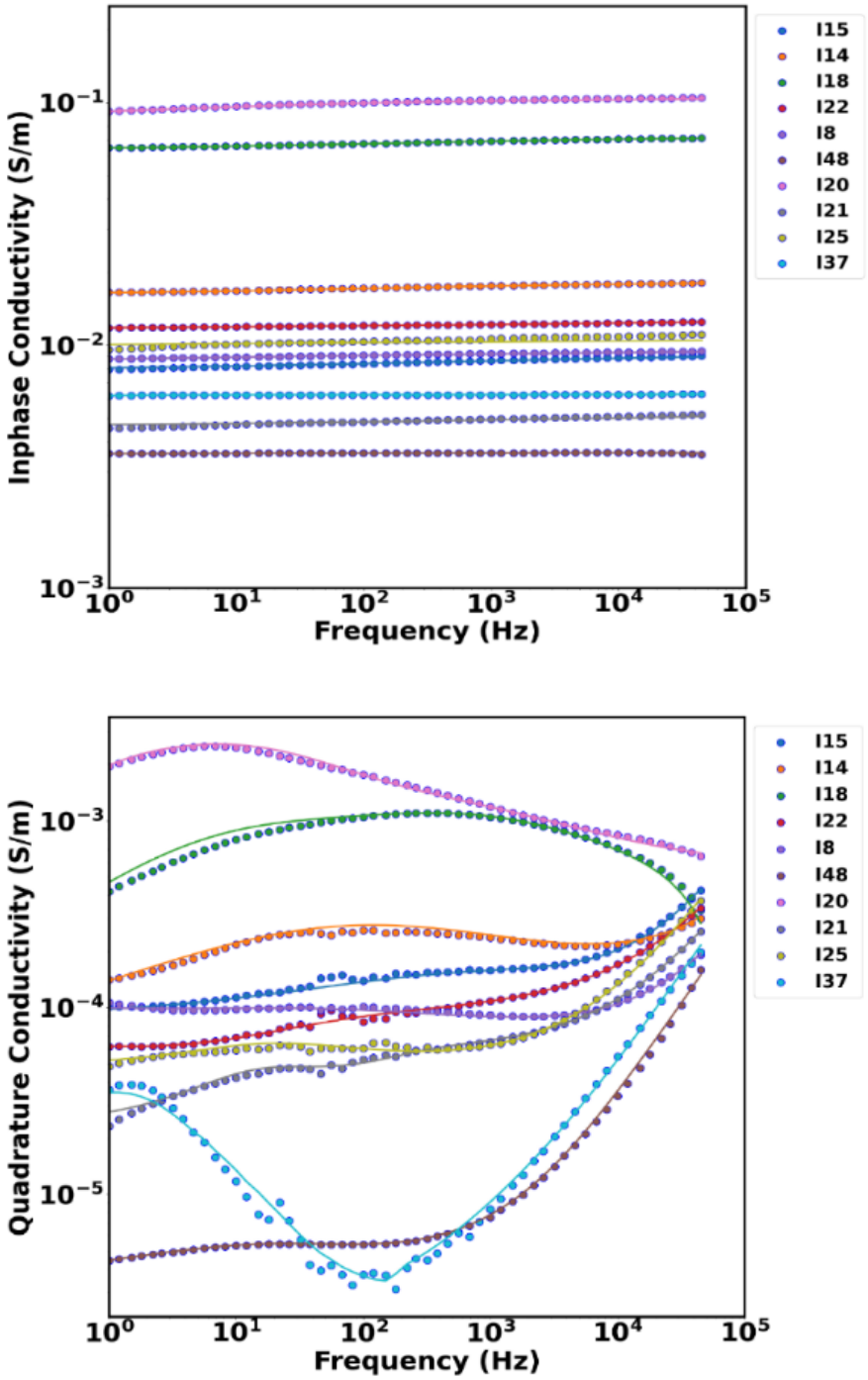


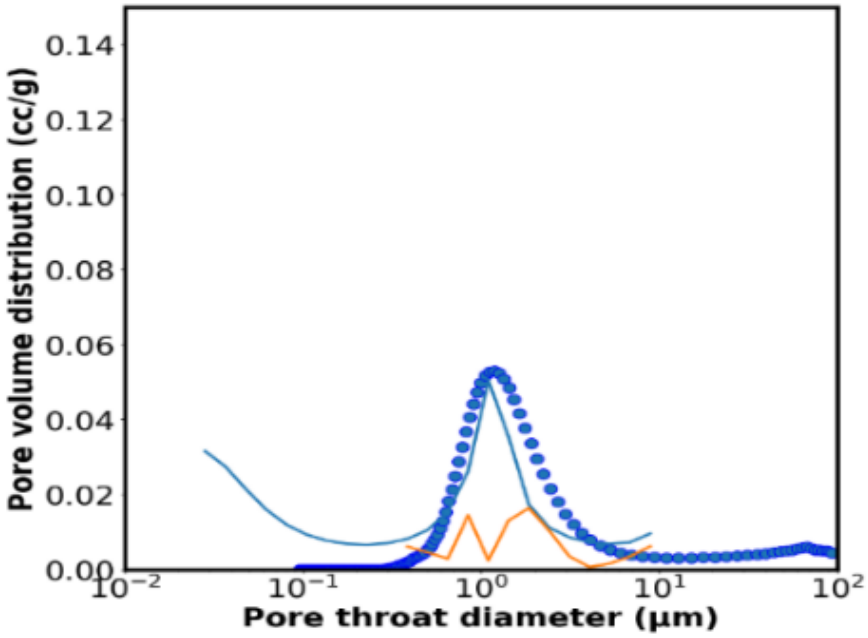
Figure 4: SIP spectra fitted with Double Cole-Cole model. Circles displays measured SIP data, while Line depicts the fitted data.

Table 1: Inverted Double Cole-Cole parameters used to fit the laboratory complex conductivity data. The first Cole-Cole model is used to represent the low-frequency dispersion, whereas the second mechanism represents low-frequency dispersion. The inversion is done using the RTD code as described in appendix A.

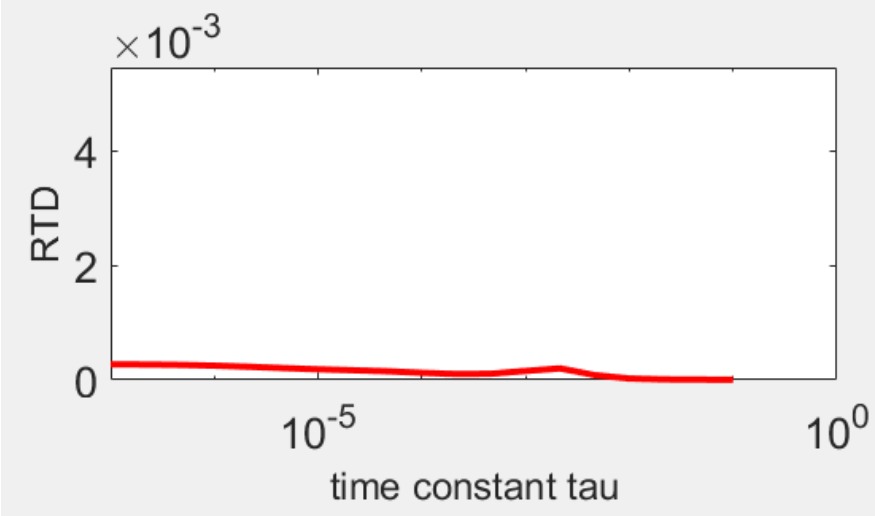
Posted on 29 Mar 2025 — The copyright holder is the author/funder. All rights reserved. No reuse without permission. — https://doi.org/10.22541/essoar.174326513.30629624/v1 — This is a preprint and has not been peer-reviewed. Data may be preliminary.

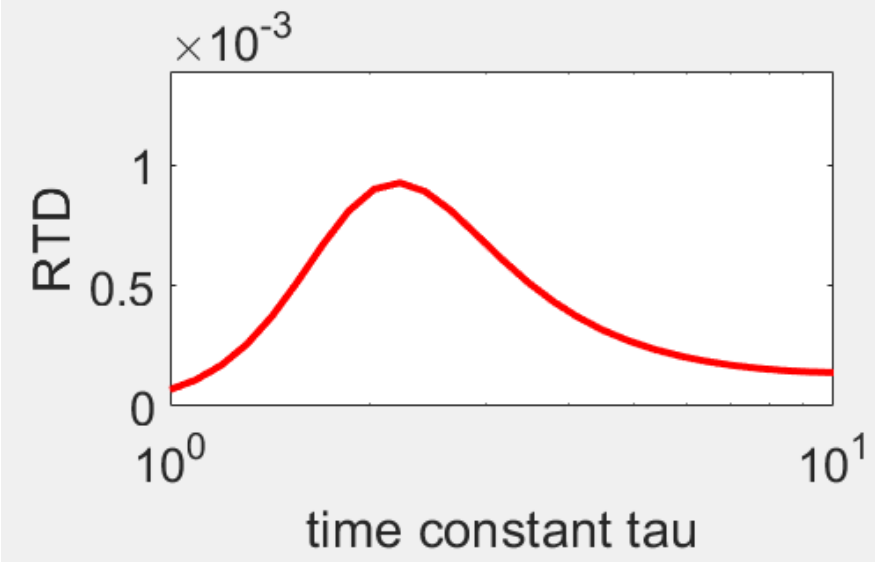
Sample ID	$\sigma_{[?]}$	M_1	M_2	c_1	c_2	τ_1	τ_2	Real/Quad error
I14	0.0184	0.0025	0.0008	0.45	0.2	3.16	0.0015	4.17E-05/1.04E-05
I18	0.073	0.0044	0.00046	0.6	0.4	10	0.0155	2E-04/9.9E-06
I22	0.0128	0.000275	0.00198	0.6	0.3	10	1E-07	1.59E-05/2.74E-06
I25	0.011	0.00010	0.0020	0.81	0.1	2.56	0.01	5E-04/1.16E-05
I8	0.0095	0.00014	0.00148	0.7	0.3	2.25	0.1	4.21E-05/1.11E-05
I15	0.0095	0.00043	0.0050	0.45	0.1	10	2E-04	5.33E-05/4.75E-06
I48	0.0036	8.86E-06	8.51E-05	0.8	0.18	17.01	0.017	1.17E-05/1.76E-06
I4	0.0074	0.00013	0.00092	0.8	0.2	2.21	0.1	7.04E-05/4.44E-06
I41	0.010	0.00090	0.00422	0.7	0.3	10	0.0028	1.43E-04/1.09E-05
I20	0.10	0.0135	0.0125	0.7	0.6	11.42	0.1	8.0E04/1.0E-04
I21	0.0059	0.000149	0.00079	0.9	0.1	3.59	1E-10	7.24E-05/1.39E-06

Figure 5(a)-(c) compares the pore size distribution inverted using SIP data with the pore size distribution measured using MIP technique, revealing a strong correlation between the two techniques. The improved fit and relaxation time provided by the Double Cole-Cole model for our carbonate samples suggest that it captures underlying complexities not adequately described by a single relaxation process.



Posted on 29 Mar 2025 — The copyright holder is the author/funder. All rights reserved. No reuse without permission. — https://doi.org/10.22541/essoar.174326513.30629624/v1 — This is a preprint and has not been peer-reviewed. Data may be preliminary.





Posted on 29 Mar 2025 — The copyright holder is the author/funder. All rights reserved. No reuse without permission. — https://doi.org/10.22541/essoar.174326513.30629624/v1 — This is a preprint and has not been peer-reviewed. Data may be preliminary.

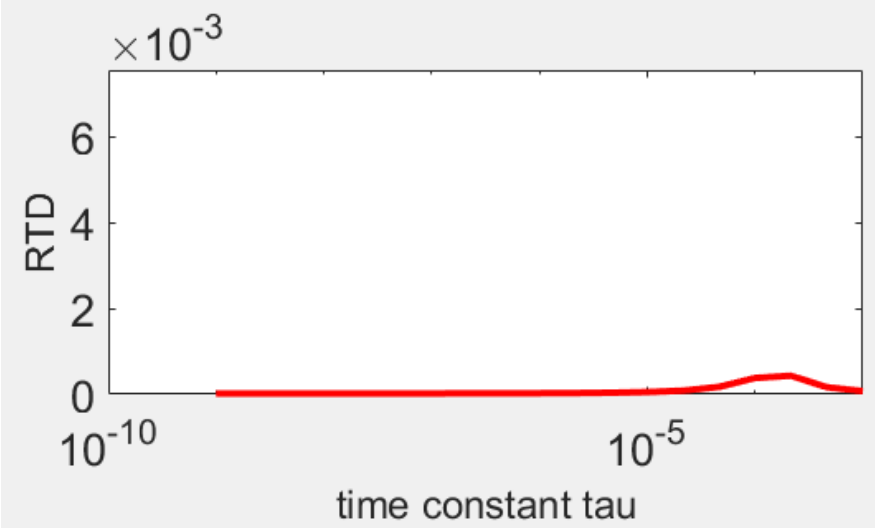
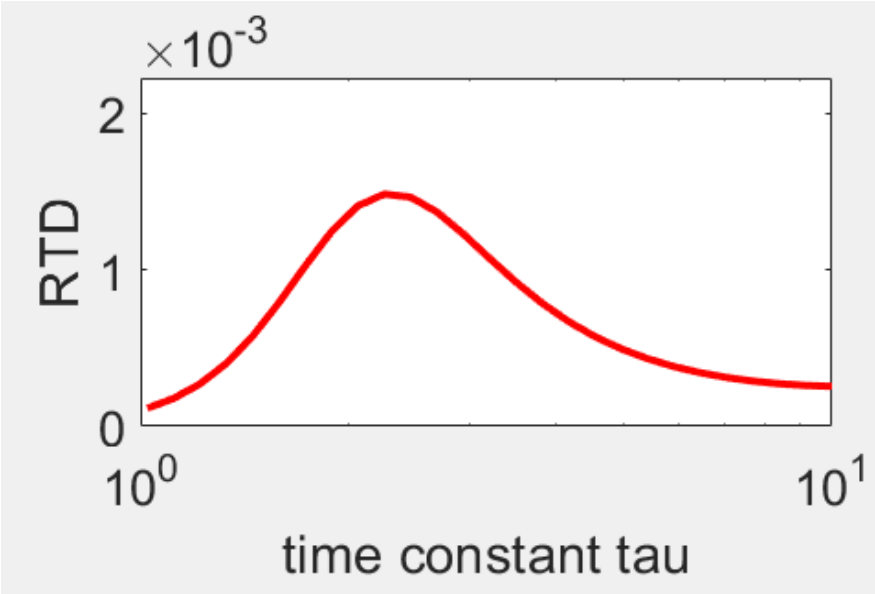
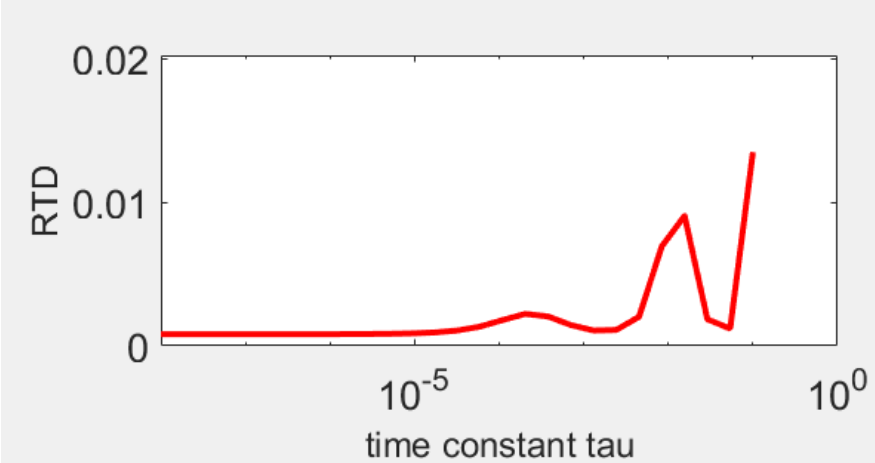
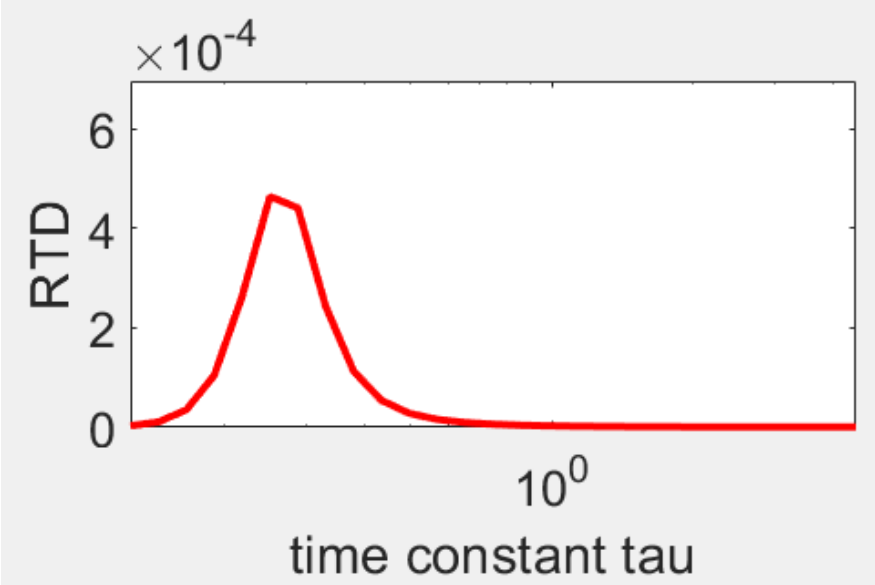


Figure 5(a): Comparison of PSD from MIP (Dotted line) and from inversion of SIP data (blue line) using RTD retrieved from RTD code (Florsch et al., 2014). The orange line shows the residual between MIP and SIP fit. The values of standard deviation, median and peak of the pore volume are given in Table 2. The RTD is given on the left side retrieved from SIP spectra inversion for each sample.



Posted on 29 Mar 2025 — The copyright holder is the author/funder. All rights reserved. No reuse without permission. — https://doi.org/10.22541/essoar.174326513.30629624/v1 — This is a preprint and has not been peer-reviewed. Data may be preliminary.





Posted on 29 Mar 2025 — The copyright holder is the author/funder. All rights reserved. No reuse without permission. — https://doi.org/10.22541/essoar.174326513.30629624/v1 — This is a preprint and has not been peer-reviewed. Data may be preliminary.

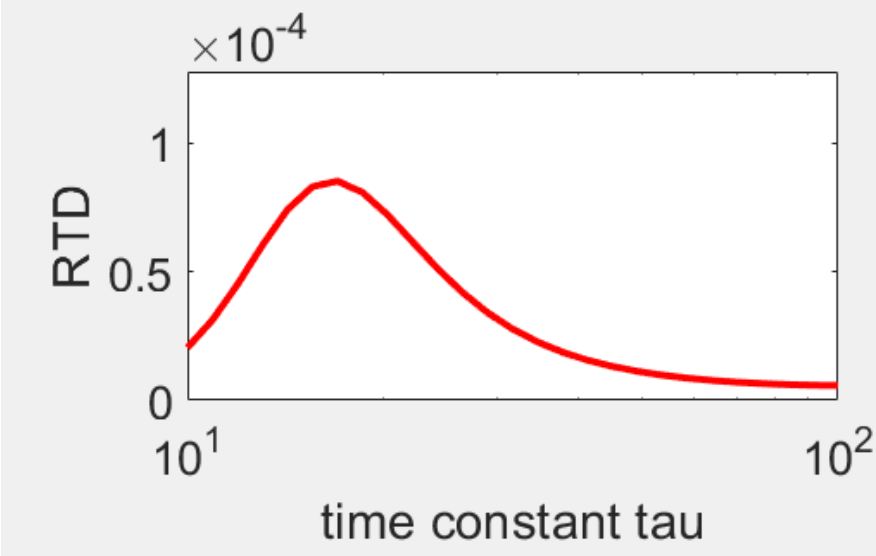
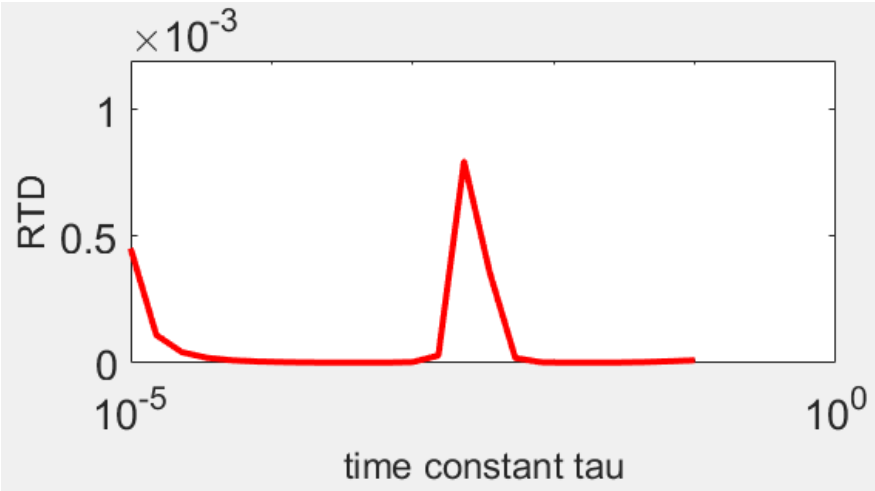


Figure 5(b): Comparison of PSD from MIP (Dotted line) and from inversion of SIP data (blue line) using RTD retrieved from RTD code (Florsch et al., 2014). The orange line shows the residual between MIP and SIP fit. The values of standard deviation, median and peak of the pore volume are given in Table 2. The RTD is given on the left side retrieved from SIP spectra inversion for each sample.



Posted on 29 Mar 2025 — The copyright holder is the author/funder. All rights reserved. No reuse without permission. — https://doi.org/10.22541/essoar.174326513.30629624/v1 — This is a preprint and has not been peer-reviewed. Data may be preliminary.

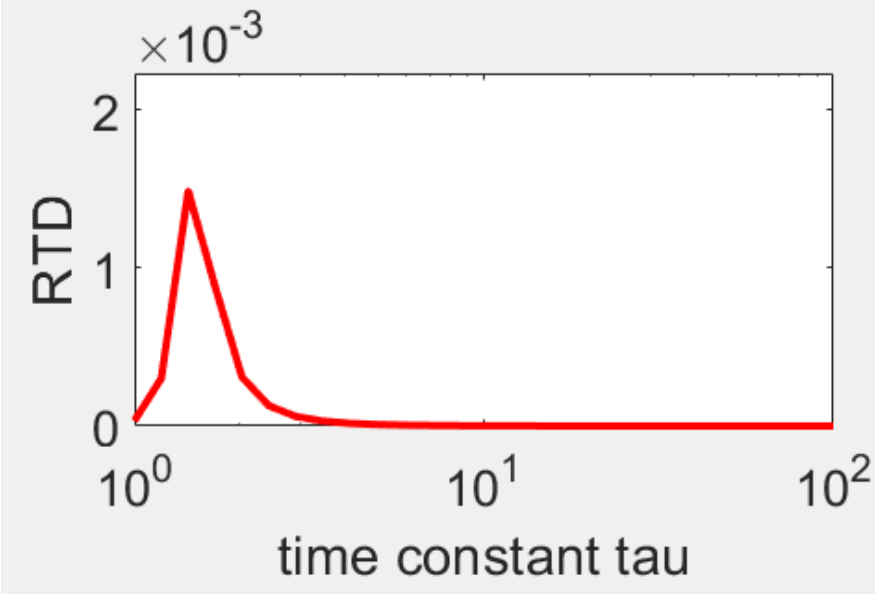


Figure 5(c): Comparison of PSD from MIP (Dotted line) and from inversion of SIP data (blue line) using RTD retrieved from RTD code (Florsch et al., 2014). The orange line shows the residual between MIP and SIP fit. The values of standard deviation, median and peak of the pore volume are given in Table 2. The RTD is given on the left side retrieved from SIP spectra inversion for each sample.

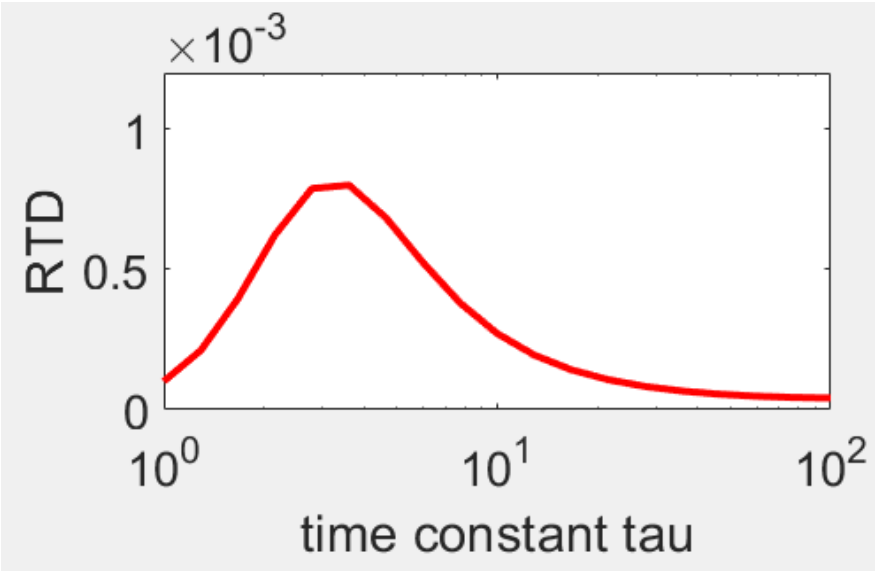


Table 2 displays the differences in the dominant peak and standard deviation of pore volume between the SIP and MIP measurements. It can be seen that the values of pore volume distributions estimated using inversion of SIP are very close to the values measured with MIP, showing the accuracy of RTD code for the carbonate samples.

Table 2. Estimated values of standard deviation, and peak of the pore volume distribution of MIP and SIP data, where 'r_c (peak) μm' is the peak value of the pore volume distribution.

Posted on 29 Mar 2025 — The copyright holder is the author/funder. All rights reserved. No reuse without permission. — https://doi.org/10.22541/essoar.174326513.30629624/v1 — This is a preprint and has not been peer-reviewed. Data may be preliminary.

Sample ID	Standard deviation (MIP data)	Standard deviation (SIP data)	r_c(peak) μm (MIP data)	r_c (peak) μm (SI
I14	0.013	0.011	0.053	0.050
I18	0.012	0.0158	0.06	0.07
I22	0.0037	0.0048	0.0139	0.0137
I25	0.0129	0.010	0.05	0.04
I8	0.0106	0.0085	0.04	0.038
I15	0.0046	0.0039	0.018	0.019
I48	0.0031	0.0024	0.01	0.009
I4	0.0074	0.0069	0.03	0.03
I41	0.0092	0.0072	0.04	0.03
I20	0.021	0.017	0.10	0.09
I21	0.0082	0.0077	0.03	0.028

The better fit and relaxation time provided by the Double Cole-Cole model for our carbonate samples suggest that it might capture some underlying complexities not adequately described by a single relaxation process. Carbonates can exhibit complex dielectric behavior due to various factors, including mineralogy, porosity, fluid content, and structural characteristics (Markov et al., 2012; Gomaa et al., 2015). Furthermore, the results were validated by SEM image analysis, where pore throat size in carbonates was estimated using 2D SEM images. The resolution of the images was determined based on the scale (magnification) at which the SEM image was acquired. The images were processed to extract the pore throats, and pore throat size distribution was estimated, as depicted in Figure 6(i)-6(v).

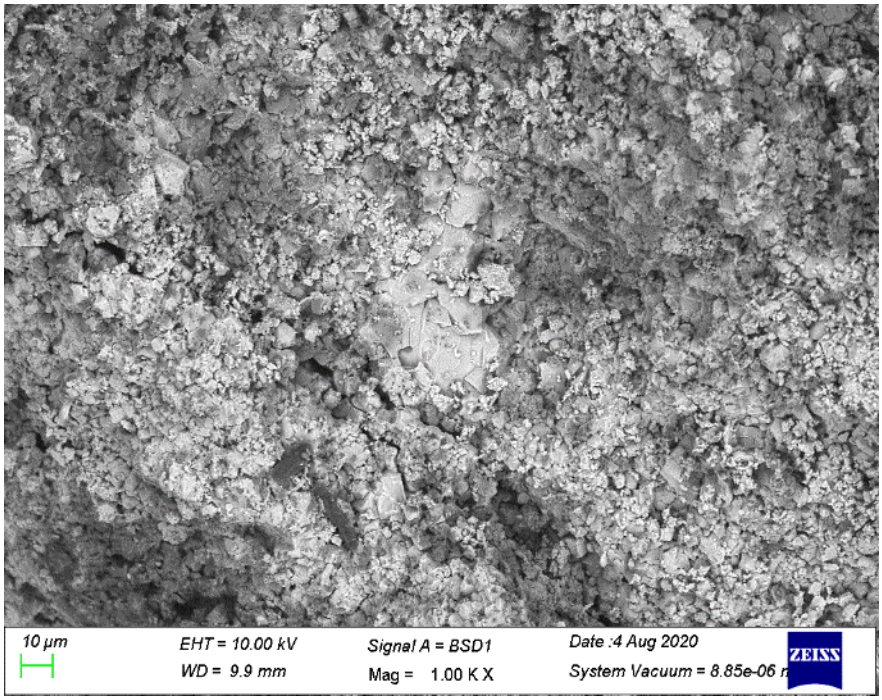


Figure 6 (i): (a & b) showing BSE SEM image (20 μm) and segmented image, respectively. (c) The White space indicates the pores while the black area represents the grains in segmented image. (d) Pore throat size distribution of sample I3 using SEM image analysis. The blue color is showing the histogram and red line is its normal distribution.

Posted on 29 Mar 2025 — The copyright holder is the author/funder. All rights reserved. No reuse without permission. — https://doi.org/10.22541/essoar.174326513.30629624/v1 — This is a preprint and has not been peer-reviewed. Data may be preliminary.

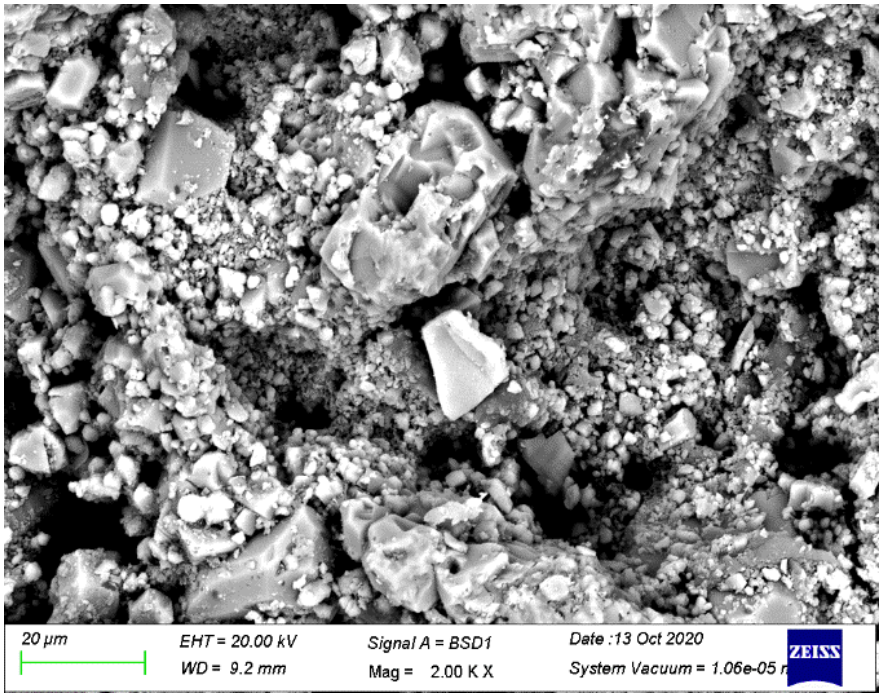


Figure 6 (ii): (a & b) showing BSE SEM image (10 µm) and segmented image, respectively. (c) The White space indicates the pores while the black area represents the grains in segmented image. (d) Pore throat size distribution of sample I14 using SEM image analysis. The blue color is showing the histogram and red line is its normal distribution.

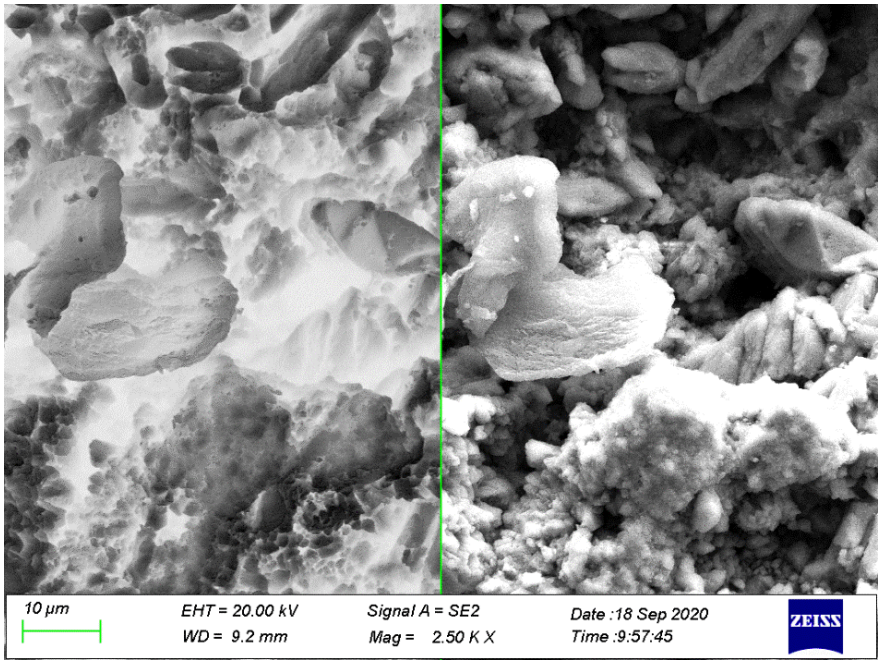


Figure 6 (iii): (a & b) showing BSE SEM image (10 µm) and segmented image, respectively. (c) The

Posted on 29 Mar 2025 — The copyright holder is the author/funder. All rights reserved. No reuse without permission. — https://doi.org/10.22541/essoar.174326513.30629624/v1 — This is a preprint and has not been peer-reviewed. Data may be preliminary.

White space indicates the pores while the black area represents the grains in segmented image. (d) Pore throat size distribution of sample I20 using SEM image analysis. The blue color is showing the histogram and red line is its normal distribution.

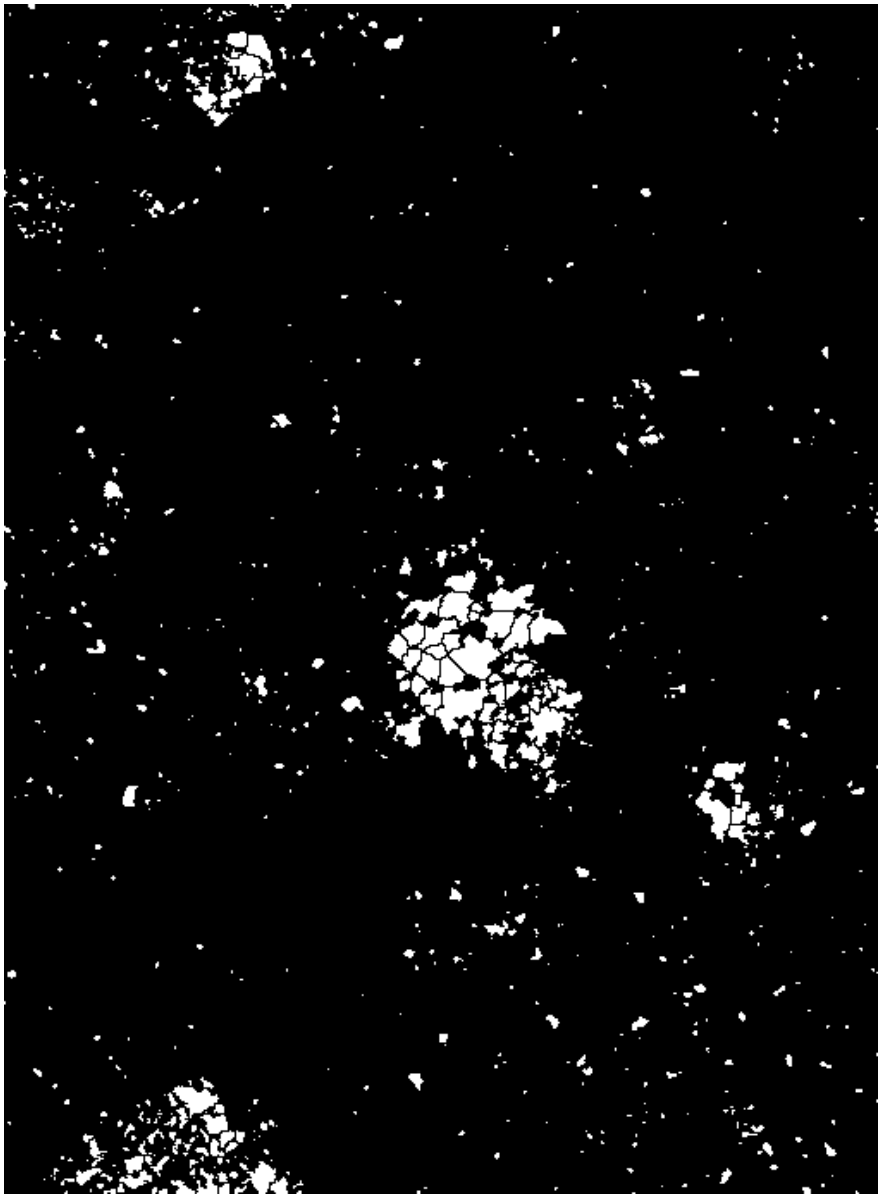


Figure 6 (iv): (a & b) showing BSE SEM image (100 μm) and segmented image, respectively (c)The White space indicates the pores while the black area represents the grains in segmented image. (d) Pore throat size distribution of sample I21 using SEM image analysis. The blue color is showing the histogram and red line is its normal distribution.

Posted on 29 Mar 2025 — The copyright holder is the author/funder. All rights reserved. No reuse without permission. — https://doi.org/10.22541/essoar.174326513.30629624/v1 — This is a preprint and has not been peer-reviewed. Data may be preliminary.

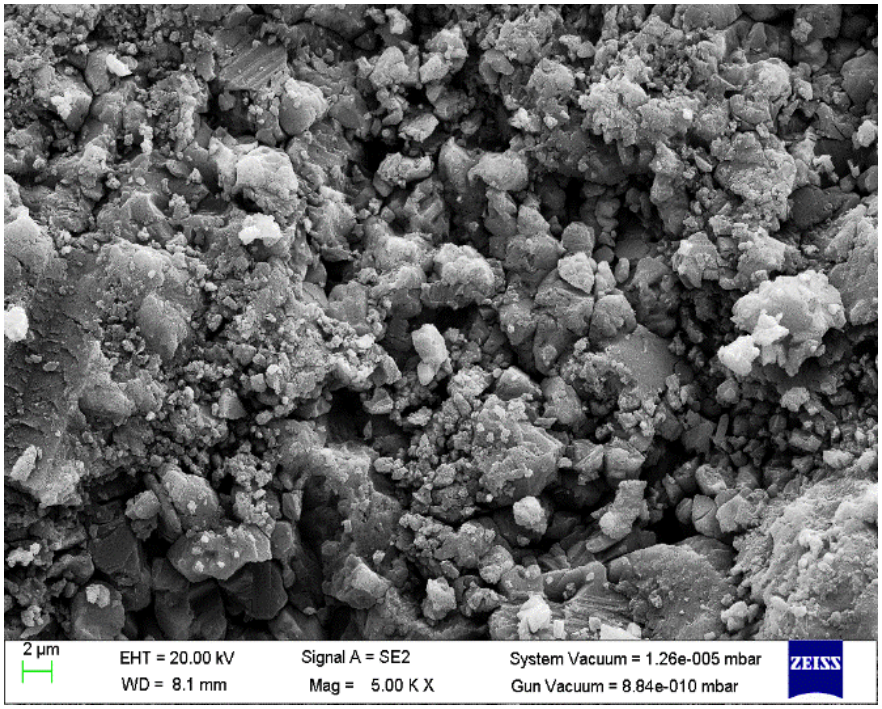


Figure 6 (v): (a & b) showing BSE SEM image (2 µm) and segmented image, respectively(c) The White space indicates the pores while the black area represents the grains in segmented image. (d) Pore throat size distribution of sample I41 using SEM image analysis. The blue color is showing the histogram and red line is its normal distribution.

The dominant peak of the pore throat size distribution was found to be in the range of 0.09-50µm, which was consistent across all three techniques i.e., SIP, MIP, and SEM. According to Lonoy (2006), microporosity is present if the pore size is less than 50 µm. This consistency across different methods reinforces the reliability of our findings, suggesting significant microporosity within the carbonate samples, which is crucial for understanding their storage and flow properties. The SEM analysis was initially conducted on a subset of representative carbonate samples. However, this analysis can be extended to include all other carbonate samples in our dataset.

4.1 Permeability model:

We conducted an investigation aimed at estimating the permeability of carbonates and gaining insights into their grain and pore size distribution. In our initial analysis, we presented data from 11 samples. However, to enhance the accuracy and comprehensiveness of our findings, we have incorporated additional samples (around 30 carbonate samples data) into Figure 7. The inclusion of these extra samples aims to provide a more robust representation of permeability, ensuring a more thorough analysis and interpretation of the results. Various models (i.e., Weller [54], Pape [55]) were tested for this purpose, and we discovered a promising model proposed by Revil et al. in 2010. This model establishes a relationship between the permeability (k), formation factor (F), quadrature conductivity (σ"), and the specific surface conductivity of the Stern layer (Σs), which is represented by Equation 14:

$$k \approx \frac{(\sum s)^2 \sigma"^{-2}}{4.5F^3} (14)$$

In agreement with this model, we set Σs to a value of 2×10ˆ-9 S as per Revil et al. (2010. Remarkably, our experimental data closely aligns with the model, as demonstrated in Figure 7. Notably, our sample set encompasses Wackestone, mudstone, and limestone.

Posted on 29 Mar 2025 — The copyright holder is the author/funder. All rights reserved. No reuse without permission. — https://doi.org/10.22541/essoar.174326513.30629624/v1 — This is a preprint and has not been peer-reviewed. Data may be preliminary.

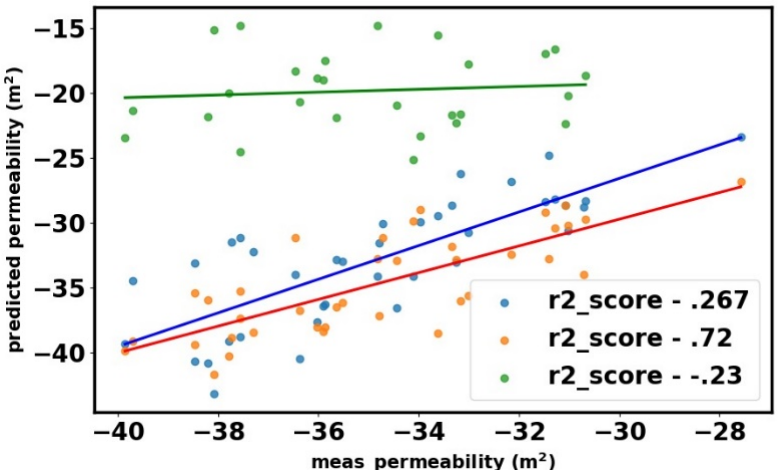


Figure 7: Comparison with the predicted and measured permeability of carbonates using different models. (Red) Revil model$k \approx \frac{(\sum s)^2 \sigma"^{-2}}{4.5F^3}$, Weller's model $K = \frac{2.66*10^{-7}}{\sigma"^{0.66}F^{5.35}}$, (Blue) and Paris model,$K = \frac{475*10^{-12}}{\text{Spor}^{3.1}F}$, (green), where σ" represents quadrature conductivity at 1Hz , F is formation factor, and Spor is normalized specific surface area."

Our analysis further unveiled a broader range of grain size distribution within our carbonate samples, a finding substantiated by the patterns displayed in Figure 8 and Figure 9. Figure 8 depicts the interdependence of quadrature conductivity with the product of formation factor and permeability, described by the relationship in equation 15:

$kF^3 \approx \left(\frac{1}{\sigma"^2}\right) \approx E_h^{-2}(15)$

Where,

$$E_h = \int_0^{+\infty} \eta \, \eta(\eta) \, |d\eta|$$

$$\eta = \frac{1}{D}$$

Here, E_h is the anticipated value of the probability density function h(η), where η is inverse of the grain diameter (D). This link bridges petrophysical properties (permeability and formation factor) with electrical attributes (electrical conductivity), suggesting that variations in grain size distribution, which influence formation factor and relaxation behavior, exert a dual impact on fluid flow and electrical responses.

Posted on 29 Mar 2025 — The copyright holder is the author/funder. All rights reserved. No reuse without permission. — https://doi.org/10.22541/essoar.174326513.30629624/v1 — This is a preprint and has not been peer-reviewed. Data may be preliminary.

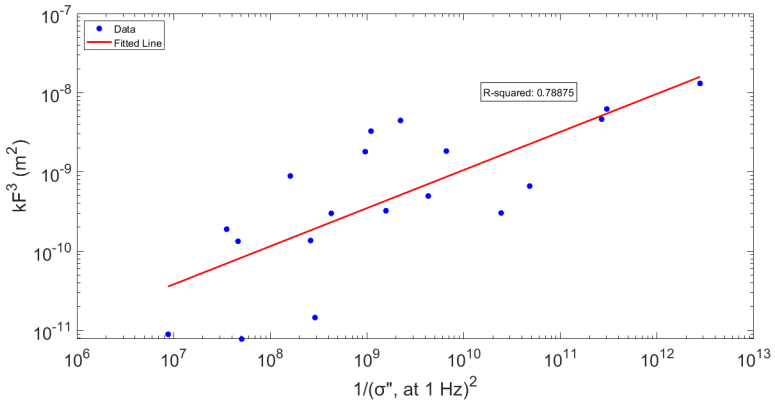


Figure 8: Relationship between quadrature conductivity and permeability along with the formation factor. The behavior of quadrature conductivity is influenced by the fluctuations in pore structure and connectivity, revealing insights into the internal organization of grains and pores.

In Figure 9, we visually present the grain size distribution in wackestone and mudstone carbonate samples. The broader grain size distribution in wackestone implies the coexistence of fine-grained carbonate mud and larger biogenic fragments within this rock type. In contrast, mudstone, typically characterized by fine-grained particles such as clay and silt, exhibits a wider grain size distribution, possibly due to factors like sedimentation rates, sediment sources, or mixing processes.

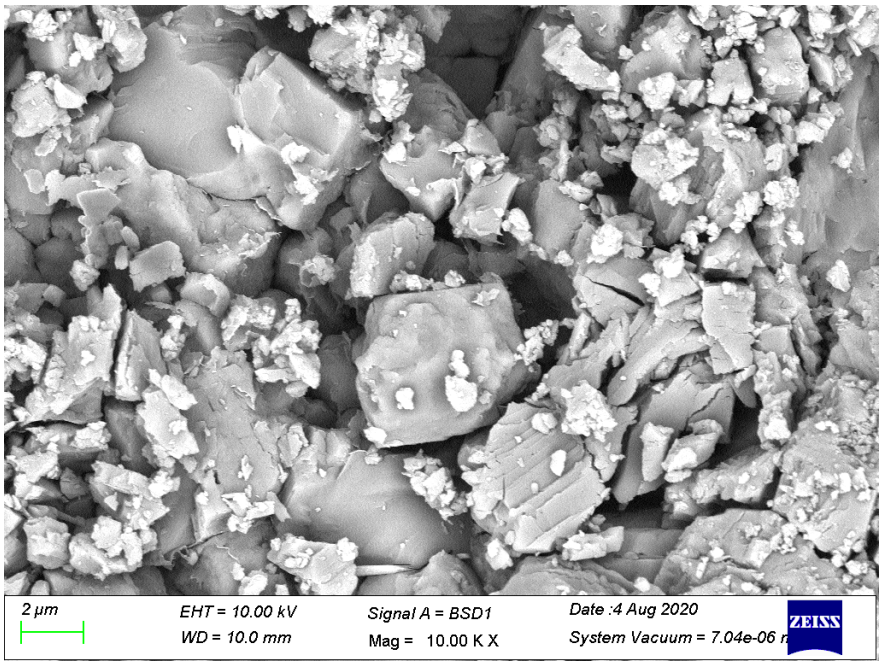


Figure 9: The grain size distribution of (a) wackestone, (b) mudstone, and (c) limestone carbonate sample using SEM image analysis. The BSE SEM images are displayed on the left corner and segmented image in the middle (from left to right).

While Revil et al. (2010) previously concluded that the approach under consideration would be effective in the presence of Stern layer polarization, our current study suggests that Stern layer polarization might

Posted on 29 Mar 2025 — The copyright holder is the author/funder. All rights reserved. No reuse without permission. — https://doi.org/10.22541/essoar.174326513.30629624/v1 — This is a preprint and has not been peer-reviewed. Data may be preliminary.

be the principal mechanism of polarization in carbonate rocks. This proposition is particularly relevant to the carbonate reservoirs of the Indian western offshore, characterized by mineralogical composition, porosity range, and textural heterogeneity typical of Tertiary-aged carbonate reservoirs found globally (Sharma et al., 2006). However, this needs to be further investigated by applying other models to determine which polarization phenomena are dominant in carbonates.

5.Conclusion:

Our comprehensive investigation embarked measurement of electrical attributes, on the estimation of permeability in carbonate rocks, and unraveling intricate insights into their grain-to-pore size distribution. Through rigorous testing of various models, we identified a robust framework that bridges quadrature conductivity, formation factor, and permeability to the specific surface conductivity of the Stern layer in the carbonate samples.

The model's alignment with our experimental data challenges the conventional assumptions and underscores its applicability across diverse carbonate rock types, ranging from wackestone to mudstone and limestone. Understanding the grain size variations in wackestone and mudstone highlights the significant role of depositional processes in shaping carbonate rock properties, which is essential for geological and reservoir studies. The RTD code successfully estimated the pore size distribution in carbonate rocks, demonstrating the suitability of the Double Cole-Cole model to account for their complex nature, thereby surpassing the limitations of the Cole-Cole model. This pivotal observation, supported by the classification of pore size distribution as microporous according to the Lonoy classification. holds paramount importance in estimating and understanding of reservoir's petrophysical properties, including storage, saturation, and permeability, which are crucial for effective reservoir management.

The Revil model demonstrated encouraging results in estimating permeability for carbonate rocks, showing good agreement with experimental data ($?^2$=0.788). However, the assumption of a constant specific surface conductivity ($\Sigma?$=2×10^{-9}S) may not fully capture variations in pore architecture and mineralogy across different carbonate lithologies. Future research should explore variable values to enhance the model's accuracy and applicability to diverse geological settings.

Our research journey not only enhanced our understanding of carbonate rock pore structures but also paved the way for future investigations. Exploring the impact of diverse parameters, such as temperature and pressure, on pore size distribution, holds the potential to further enrich our comprehension of carbonate rock systems and their intricate interplay with fluid transport and electrical conductivity using non-destructive methods like SIP.

Acknowledgement:

We express our sincere gratitude for the invaluable support and encouragement received from the Rock and Fluid Multiphysics Group, Department of Earth Sciences, IIT Roorkee. The data pivotal to this study were graciously provided by Oil and Natural Gas Corporation Limited (ONGC), India. Special thanks to Mr. S.S. Khanna and Mr. A. Ghosh from ONGC Dehradun for their insightful guidance and mentorship, which significantly enriched the research process.

We extend our appreciation to Mr. S.K. Behera from the Central Institute of Mining and Fuel Research, India, for his instrumental role in conducting mercury intrusion porosimetry (MIP) measurements.

A heartfelt acknowledgment is also owed to Mr. Gagan Choudhary, Head Gemmologist, and Director from IIGJ-Research & Laboratories Centre, Jaipur, for his unwavering support in the scanning of the samples.

Author Contributions:

N. Panwar: Methodology, Investigation, Writing-original draft

R. Sharma: Methodology, Investigation, Conceptualization, Supervision, Writing-original draft

H.J. Kalita: Writing- review & editing

Posted on 29 Mar 2025 — The copyright holder is the author/funder. All rights reserved. No reuse without permission. — https://doi.org/10.22541/essoar.174326513.30629624/v1 — This is a preprint and has not been peer-reviewed. Data may be preliminary.

A. Dhiman: Coding- editing, Writing- review & editing

P. Soni: Coding- editing, Writing- review & editing

S. Malik: Writing-review & editing

P. Pandey: Writing- review & editing

Statement of Data and Code Availability:

Data and code would be shared if a request is received for the same.

Competing Interests:

The author declare that they have no competing interests.

References:

Abell, A.B., Willis, K.L., Lange, D.A., (1999) Mercury intrusion porosimetry and image analysis of cement-based materials. J. Colloid Interface Sci. 211, 39–44.

Beiranvand, Bijan. "Quantitative Characterization of Carbonate Pore Systems by Mercury- Injection Method and Image Analysis in a Homogeneous Reservoir." Paper presented at the Middle East Oil Show, Bahrain, June 2003. doi: https://doi.org/10.2118/81479-MS.

Bohnsack, D., Potten, M., Freitag, S., Einsiedl , F., & Zosseder, K. (2021). Stress sensitivity of porosity and permeability under varying hydrostatic stress conditions for different carbonate rock types of the geothermal Malm reservoir in Southern Germany. Geothermal Energy. doi:https://doi.org/10.1186/s40517-021-00197-w.

Brown, E. (2011). Successfully Stimulating Complex Carbonate Reservoirs.*Society of Petroleum Engineers.*

Bust, V. K., Oletu, J. U., & Worthington, P. F. (2011). The Challenges for Carbonate Petrophysics in Petroleum Resource Estimation.*Society of Petroleum Engineers.*

Cao, Z., Liu, G., Zhan, H., Li, C., You, Y., Yang, C., & Jiang, H. (2016). Pore structure characterization of Chang-7 tight sandstone using MICP combined with N2GA techniques and its geological control factors.*Scientific Reports.* doi: 10.1038/srep36919.

Choquette, P.W. and Pray, L.C. (1970) Geologic Nomenclature and Classification of Porosity in Sedimentary Carbonates. American Association of Petroleum Geologists Bulletin, 54, 207-250.Cnudde, V., Cwirzen, A., Masschaele, B., & Jacobs, P. J. (2009). Porosity and microstructure characterization of building stones and concretes.*Engineering Geology*, 76–83. doi:10.1016/j.enggeo.2008.06.014.

Cnudde, V., Cwirzen, A., Masschaele, B., & Jacobs, P. J. (2009). Porosity and microstructure characterization of building stones and concretes. Engineering Geology, 76–83. doi:10.1016/j.enggeo.2008.06.014.

Coperey, A., Revil, A., Abdulsamad, F., Stutz, B., Duvillard, P. A., & Ravanel, L. (2019). Low-Frequency Induced Polarization of Porous Media Undergoing Freezing: Preliminary Observations and Modeling. *JGR Solid Earth*, 12. doi:10.1029/2018JB017015.

Cunningham, K. J., Sukop, M., & Curran, H. A. (2012). Carbonate Aquifers. In *Developments in Sedimentology* (pp. 869-896). doi:10.1016/B978-0-444-53813-0.00028-9.

Diamond, S., Leeman, M.E. Pore Size Distributions in Hardened Cement Paste by Sem Image Analysis. MRS Online Proceedings Library 370, 217–226 (1994). https://doi.org/10.1557/PROC-370-217.

Florsch, N., Revil, A., & Camerlynck, C. (2014). Inversion of generalized relaxation time distributions with optimized damping parameter. Journal of Applied Geophysics, 119-132.

Gomaa, M. M., Kassab, M. A., & El-Sayed, N. A. (2015). Study of pertrographical and electrical properties of some Jurassic carbonate rocks, north Sinai, Egypt. Egyptian Journal of Petroleum, 343-352. doi:https://doi.org/10.1016/j.ejpe.2015.07.003.

Posted on 29 Mar 2025 — The copyright holder is the author/funder. All rights reserved. No reuse without permission. — https://doi.org/10.22541/essoar.174326513.30629624/v1 — This is a preprint and has not been peer-reviewed. Data may be preliminary.

Gonzalez, R.C., & Woods, R.E. (2018). Digital Image Processing, Fourth Edition. Pearson Education Limited, 786-793, ISBN 978-0-13-335672-4.

Halisch, M., Weller, A., Hupfer, S., & Kassab, M. A. (2014). Impedance Spectroscopy on Carbonates. *Annual Symposium of the Society of Core Analysts.* Avignon, France. doi:10.13140/2.1.3079.9045.

Hussein, D., Lawrence, J., Rashid, F., Glover, P., & Lorinczi, P. (2018). Engineering in Chalk. London-UK. doi:10.1680/eiccf.64072.529.

Irajian, A. A., Guilani, K. B., Mahari, R., & Solgi, A. (2016). Porosity and Rock-Typing in Hydrocarbon Reservoirs Case Study in Upper Member of Dalan Formation in Kish Gas Field, South of Zagros, Iran. Open Journal of Geology, 399-409. doi:10.4236/ojg.2016.66035.

Jougnot, D., Ghorbani, A., Revil, A., Leroy, P., & Cosenza, P. (2010). Spectral induced polarization of partially saturated clay-rocks: a mechanistic approach. Geophysical JournalInternational, 210–224. doi:https://doi.org/10.1111/j.1365-246X.2009.04426.x.

Kargarpour, M. A. (2020). Carbonate reservoir characterization: an integrated approach. *Journal of Petroleum Exploration and Production Technology*, 2655–2667.

Kovacs, A. (2021). Quantitative classification of carbonate aquifers based on hydrodynamic behaviour. *Hydrogeology Journal volume*, 33-52.

Li, C., Chen, G., Qiu, X., Lou, Q., & Gao, X. (2021). A direct proof for Maxwell–Wagner effect of heterogeneous interface. AIP advances. doi: 10.1063/5.0040947.

Liu, J., Song, S., Cao, X., Meng, Q., Pu, H., Wang, Y., & Liu, J. (2020). Determination of full-scale pore size distribution of Gaomiaozi bentonite. Journal of Rock Mechanics and Geotechnical Engineering, 403-413. doi:https://doi.org/10.1016/j.jrmge.2019.12.005.

Liu, X., Kong, L., Zhou, K. (2014). A time domain induced polarization relaxation time spectrum inversion method based on a damping factor and residual correction. Pet. Sci. 11, 519–525. https://doi.org/10.1007/s12182-014-0367-9.

Lonoy, A., 2006, Making sense of carbonate pore systems: AAPG Bulletin, v. 90/9, p. 1381-1405.

Lonoy, A., 2006, Making sense of carbonate pore systems: American Association of Petroleum Geologists (AAPG) Bulletin, 90, 1381-1405. doi:10.1306/03130605104.

Lu, H., Tang, H., Wang, M., Li, X., Zhang, L., Wang, Q., Zhao, Y., Zhao, Y., Liao, J. (2021). Pore Structure Characteristics and Permeability Prediction Model in a Cretaceous Carbonate Reservoir, North Persian Gulf Basin. *Geofluids.* doi:10.1155/2021/8876679.

Lu, H., Tang, H., Wang, M., Li, X., Zhang, L., Wang, Q., Zhao, Y., Zhao, Y., Liao, J. (2021). Pore Structure Characteristics and Permeability Prediction Model in a Cretaceous Carbonate Reservoir, North Persian Gulf Basin. Geofluids. doi:10.1155/2021/8876679.

Lucia, F. J. (1983). Petrophysical Parameters Estimated From Visual Descriptions of Carbonate Rocks: A Field Classification of Carbonate Pore Space. Journal of Petroleum Technology, 629–637. doi:https://doi.org/10.2118/10073-PA.

Lucia, F. J. (1995). Rock-Fabric/Petrophysical Classification of Carbonate Pore Space for Reservoir Characterization. AAPG Bulletin, 1275–1300. doi:https://doi.org/10.1306/7834D4A4-1721-11D7-8645000102C1865D.Lucia, F. J. (2007). Carbonate Reservoir Characterization. Berlin, Heidelberg: Springer. doi:https://doi.org/10.1007/978-3-540-72742-2.

Markov, M., Kazatchenko, E., Mousatov, A., Pervago, E., 2012. The dielectric permittivity of carbonate formations from the unified microstructure model. Journal of Applied Geophysics 76, 56–63.

Posted on 29 Mar 2025 — The copyright holder is the author/funder. All rights reserved. No reuse without permission. — https://doi.org/10.22541/essoar.174326513.30629624/v1 — This is a preprint and has not been peer-reviewed. Data may be preliminary.

Muller-Huber, E., B"orner, F., B"orner, J.H. & Kulke, D., 2018. Combined interpretation of NMR, MICP, and SIP measurements on mud-dominated and grain-dominated carbonate rocks, J. Appl. Geophys., 159, 228–240.

Niu, Q., & Zhang, C. (2018). Joint inversion of NMR and SIP data to estimate pore size distribution of geomaterials. Geophysics Journal International, 1791–1805. doi:10.1093/gji/ggx501.

Osterman, G., Keating, K., Binley, A., and Slater, L. (2016), A laboratory study to estimate pore geometric parameters of sandstones using complex conductivity and nuclear magnetic resonance for permeability prediction, Water Resour. Res., 52, 4321– 4337, doi:10.1002/2015WR018472.

Pape, H., Riepe, L., & Schopper, J. (1987). Theory of self-similar network structures in sedimentary and igneous rocks and their investigation with microscopical and physical method. Journal of Microscopy, 148, 121–147.

Pelton, W. H., Ward, S. H., Hallof, P. G., Sill, W. R., & Nelson, P. H. (1978). Mineral Discrimination and Removal of Inductive Coupling with Multifrequency IP. Geophysics, 588-609.

Revil, A., & Florsch, N. (2010). Determination of permeability from spectral induced polarization in granular media. Geophysical Journal International, 1480–1498. doi:10.1111/j.1365-246X.2010.04573.x.

Revil, A., Coperey, A., Shao, Z., Florsch, N., Fabricius, I. L., Deng, Y., Delsman, J. R., Pauw, P.S., Karaoulis, M., Louw, P.G.B. de., Baaren, E. S. van., Dabekaussen, W., Menkovic, A., Gunnink, J. L. (2017). Complex conductivity of soils. Water Resources Research, 7121–7147. doi:10.1002/2017WR020655.

Revil, A., Florsch, N., & Camerlynck, C. (2014). Spectral induced polarization porosimetry. *Geophysical Journal International*, 1016–1033. doi:https://doi.org/10.1093/gji/ggu180.

Revil, A., Koch, K., & Holliger, K. (2012). Is it the grain size or the characteristic pore size that controls the induced polarization relaxation time of clean sands and sandstones? *Water ResourceResearch*, W05602. doi:10.1029/2011WR011561.

Safari, H., Balcom, B. J., & Afrough, A. (2021). Characterization of pore and grain size distributions in porous geological samples – An image processing workflow. Computers & Geosciences, 104895. doi:https://doi.org/10.1016/j.cageo.2021.104895.

Sharma, R., Prasad, M., Batzle, M., & Vega, S. (2013). Sensitivity of flow and elastic properties to fabric heterogeneity in carbonates. Geophysical Prospecting, 270–286. doi: 10.1111/1365- 2478.12030.

Sharma, R., Prasad, M., Surve, G. and Katiyar, G.C., 2006, October. On the applicability of Gassmann model in carbonates. In SEG International Exposition and Annual Meeting (pp. SEG-2006). SEG.

Stingaciu, L. R., Weihermuller, L., Pohlmeier, S. H., Stapf, S., Vereecken, H., & Pohlmeier, A. (2010). Determination of pore size distribution and hydraulic properties using nuclear magnetic resonance relaxometry: A comparative study of laboratory methods. Water Resources Research, W11510. doi:10.1029/2009WR008686.

Sun, B., Yang, Q., Zhu, J., Shao, T., Yang, Y., Hou, C., & Li, G. (2020). Pore size distributions and pore multifractal characteristics of medium and low rank coals. Scientific Reports. doi:https://doi.org/10.1038/s41598-020-79338-3.

Tarantola, A., & Valette, B. (1982). Generalized Nonlinear Inverse Problems Solved Using the Least Squares Criterion. Reviews of Geophysics and Space Physics, 219–232.

Tong, M., Li, L., Wang, W., & Jiang, Y. (2006). Determining capillary-pressure curve, pore-size distribution, and permeability from induced polarization of shaley sand. Geophysics, N33 N40. doi:10.1190/1.2195989.

Vinegar, H. J., & Waxman, M. H. (1988). *United States Patent No. 4,743,854.*

Washburn, E. W. (1921). Note on a Method of Determining the Distribution of Pore Sizes in a Porous Material. Proceedings of the National Academy of Sciences, PNAS, 115-116. doi:10.1073/pnas.7.4.115.

Posted on 29 Mar 2025 — The copyright holder is the author/funder. All rights reserved. No reuse without permission. — https://doi.org/10.22541/essoar.174326513.30629624/v1 — This is a preprint and has not been peer-reviewed. Data may be preliminary.

Weiqiang, L., Longxin, M., Lun, Z., Jianxin, L., Shuqin, W., Zifei, F., Dali, S.,Changhai, L., Fachao, S., Wenqi, Z., Meng, S. (2020). Pore-throat structure characteristics and its impact on the porosity and permeability relationship of Carboniferous carbonate reservoirs in eastern edge of Pre-Caspian Basin. *Petroleum Exploration And*, 1027–1041.

Weller, A., Slater, L., Binley, A., Nordsiek, S., & Xu, S. (2015a). Permeability prediction based on induced polarization: Insights from measurements on sandstone and unconsolidated samples spanning a wide permeability range. Geophysics, 80, D161–D173.

Widiatmoko, E., Abdullah, M., & Khairurrijal. (2010). A Method to Measure Pore Size Distribution of Porous Materials Using Scanning Electron Microscopy Images. AIP Conference Proceedings, (p. 1284). doi:https://doi.org/10.1063/1.3515554.

Yao, J., Hu, R., Wang, C., & Yang, Y. (2014). Multiscale Pore Structure Analysis in Carbonate Rocks. International Journal for Multiscale Computational Engineering. doi: 10.1615/IntJMultcompEng.2014010500.

Zhang, F., Q. Niu, and C. Zhang, (2018), Estimating pore size distribution in carbonate reservoir rocks using joint inversion of NMRand complex conductivity data: SEG International Exposition and 88th Annual Meeting.

Zhang, Z., Kruschwitz, S., Weller, A., & Halisch, M. (2018). Solid Earth, Enhanced pore space analysis by use of μ-CT, NMR, MIP and SIP. 1225–1238. doi:https://doi.org/10.5194/se-9-1225-2018.

Zheng, S., Yao, Y., Elsworth, D., Wang, B., & Liu, Y. (2020). A novel pore size classification method of coals: Investigation based on NMR relaxation. Journal of Natural Gas Science and Engineering. doi:https://doi.org/10.1016/j.jngse.2020.103466.

Zimmermann, E., Kemna, A., Berwix, J., Glaas, W., Munch, H. M., & Huisman, J. A. (2008). A high-accuracy impedance spectrometer for measuring sediments with low polarizability. Measurement Science and Technology, 9.

Appendix A:

We utilized the open-source MATLAB code of Cole-Cole model to extend it for Double Cole-Cole model for complex pore size distribution in carbonate rocks. As we know that Double Cole-Cole equation involves one more term than the Cole-Cole equation which makes more parameters to estimate using inversion. So, we have followed the steps of inversion skeleton.